

VINAY KUMAR

Hyderabad, India · +91-83099-76969 · vickiease@gmail.com · [LinkedIn](#) · [GitHub](#)

EDUCATION

M.Tech, Software Systems — BITS Pilani, Dubai Campus

2026 – 2028 (Expected)

CGPA: In Progress · Coursework: Advanced Distributed Systems, Cloud Architecture

B.Tech, Computer Science & Engineering (AI) — VIT-AP University

Jun 2021 – Jul 2025

CGPA: 8.0 · Coursework: OS, Database Systems, Computer Networks, Distributed Systems

- S Grade (Top Distinction) in DSA; solved 300+ LeetCode problems (Rating 1850+) — graphs, DP, trees.

EXPERIENCE

Software Engineer (Part-time, Remote — Concurrent with M.Tech)

May 2026 – Present

micro1

- Reduced average feature delivery time by ~30% (10→7 days) by introducing system design reviews, API contract mocking, and implementation planning across a Java/Spring Boot, React, and PostgreSQL microservices platform; sole owner of features from schema design and API contract through production deployment and post-release monitoring within an 8-person team.
- Proposed and implemented an event-driven async notification architecture, reducing p99 API latency by ~40% by decoupling notification delivery from the critical request path.
- Improved engineering workflow reliability by cutting average deployment cycle time by ~25% through CI/CD pipeline enhancements, observability tooling integration, and deployment process standardisation.

Software Engineer Intern — Alethe Labs

Jun 2025 – Aug 2025

Alethe Labs India Pvt Ltd. · Gurgaon, Haryana, India

Python · FastAPI · React

- Consolidated 8 config endpoints into a unified admin platform serving ~10,000 daily requests, eliminating ~35% of inbound support tickets caused by misconfiguration.
- Diagnosed root cause of UI desync bugs — components fetched stale data because cache keys ignored query params, causing silent overwrites across concurrent requests; redesigned the data-fetching layer with scoped cache invalidation, reducing bug reproductions by ~80% (5→1 per sprint).

PROJECTS

Distributed Job Scheduling Platform

github.com/Vcode2407/distributed-job-scheduler

Java 21 · Spring Boot 3 · Kafka · PostgreSQL · Redis · React · Testcontainers · Docker

- Designed a distributed job scheduler supporting immediate, delayed, and cron-based execution with at-least-once delivery guarantees across isolated queues.
- Engineered concurrency via PostgreSQL FOR UPDATE SKIP LOCKED, Redis distributed locks, worker leasing, and lease-expiry recovery to coordinate multi-node execution and prevent duplicate processing.
- Optimized throughput via batch leasing, indexed due-job scans, and Kafka fan-out pipelines; validated >100,000 jobs/day under load. Built fault-tolerance: Kafka outbox, exponential backoff retries, DLQs, crash recovery, and lease reclamation (Prometheus/OpenTelemetry).

Multi-Agent LLM System for Citizen Service Routing

github.com/Vcode2407/multi-agent-citizen-routing-platform

Azure · Semantic Kernel · AKS · Kafka · MongoDB

- Built a citizen service routing system to replace a manual triage process that mis-routed ~32% of queries; orchestrated 5 LLM agents on AKS/Kafka sustaining 1,200+ req/min, 99.2% uptime, and sub-300ms p95 latency.
- Achieved 95%+ task accuracy across 10,000+ queries vs. 68% single-agent baseline (+27pp); used Azure App Insights to resolve recurring reasoning-chain failures.

OPEN SOURCE CONTRIBUTIONS

OpenTelemetry Java (Apache OpenTelemetry)

github.com/open-telemetry/opentelemetry-java/pull/8480

- Fixed baggage header parser recovery to preserve valid members on invalid percent-encoded entries; added regression tests across Linux, macOS, and Windows CI.

fmt — C++ Formatting Library (23.6k+ GitHub Stars)

github.com/fmtlib/fmt/pull/4813/commits

- Restored correctness of floating-point hex formatting on 32-bit (i686) builds in a 23.6k-star library used across production systems worldwide — root cause: fallback uint128 operator- truncated to 32 bits, corrupting the high word in format_hexfloat; diagnosed, fixed, and validated across architectures.

LEADERSHIP & ACHIEVEMENTS

SRMJEE Entrance Examination — AIR 7

2026

- Ranked AIR 7 nationally (top 0.01%) among 50,000+ candidates in a national postgraduate entrance examination.

President, Edutech Club — VIT-AP University

2024 – 2025

- Designed and delivered a 12-session curriculum (systems design, placement prep, AI); educational content used by 500+ members.

HackAura Hackathon — 2nd Place (Repository Intelligence Platform)

Java · Spring Boot · Neo4j · PostgreSQL · LLMs · Vector Database

- Indexed 50,000+ code files via semantic search, RAG, and Neo4j knowledge graph traversal for natural-language repository queries.

TECHNICAL SKILLS

Languages: Java, Python, C++, Go, Kotlin, JavaScript, TypeScript, SQL, Bash

CS Fundamentals: Data Structures & Algorithms, System Design, Software Design, Object-Oriented Design

Backend & Distributed: Distributed Systems, Concurrency, Multi-threading, Synchronization, Spring Boot, Kafka, Microservices, REST APIs, FastAPI, Stream Processing, Scalable Engineering, Production Systems, High Availability, Performance Optimization

Databases: PostgreSQL, MongoDB, Redis, Data Modeling

Cloud & DevOps: AWS (S3, Lambda, CloudWatch), Azure (AKS, Service Bus, App Insights), Kubernetes, Docker, GitHub Actions, CI/CD

Testing & Reliability: Unit Testing, Integration Testing, Testcontainers, Debugging, Observability, OpenTelemetry, Prometheus, Grafana

AI / ML: LLMs, Transformers, RAG, Information Retrieval, Fine-tuning, LLM Evaluation, CodeBERT, Semantic Kernel

Certifications: AWS Certified Developer – Associate | Databricks Certified Data Engineer